

## Excel Coding Challenge

### Problem Statement: *Recent Election Analysis*

You are a Data Analyst intern working for an election insights firm. You are given a dataset of a recent state election in India. Your task is to analyze voting patterns, party performance, and constituency-level insights using **Excel only**.

### Dataset (Copy into Excel)

Generate 100 synthetic records using chat gpt the features are given below with few records.

Constituency,State,Party,Candidate,Votes,Total\_Votes,Gender,Age  
Bangalore South,Karnataka,ABC,Ravi Kumar,45231,120000,Male,45  
Bangalore South,Karnataka,XYZ,Sneha Reddy,50321,120000,Female,38  
Bangalore South,Karnataka,PQR,Amit Shah,18448,120000,Male,52  
Mysore,Karnataka,ABC,Kiran Gowda,62341,150000,Male,41  
Mysore,Karnataka,XYZ,Divya Sharma,54321,150000,Female,36  
Mysore,Karnataka,PQR,Rahul Verma,33211,150000,Male,48  
Tumkur,Karnataka,ABC,Suresh Naik,41234,110000,Male,50  
Tumkur,Karnataka,XYZ,Anita Desai,45210,110000,Female,39  
Tumkur,Karnataka,PQR,Manoj Singh,23556,110000,Male,44  
Hubli,Karnataka,ABC,Prakash Patil,71234,180000,Male,55  
Hubli,Karnataka,XYZ,Nisha Kulkarni,65432,180000,Female,42  
Hubli,Karnataka,PQR,Rajeev Nair,43334,180000,Male,47

### Your Tasks

#### ◆ Task 1: Data Cleaning

- Check for duplicates
- Ensure numeric columns are formatted correctly
- Validate if Total Votes are consistent per constituency

#### ◆ Task 2: Vote Share Calculation

Create a new column:

**Vote % = (Votes / Total Votes) \* 100**

### ◆ Task 3: Find Winners

- Identify the **winning candidate in each constituency**
- Show:
  - Candidate Name
  - Party
  - Votes

Use:

- MAX()
- IF()
- or **Pivot Table**

### ◆ Task 4: Party Performance

Answer:

- Which party got the **highest total votes statewide?**
- Which party won the **most constituencies?**

Use:

- Pivot Tables
- SUM aggregation

### ◆ Task 5: Gender Analysis

- Total votes received by:
  - Male candidates
  - Female candidates

Insight:

Is there any gender dominance?

#### ◆ Task 6: Age Analysis

- Average age of winning candidates
- Youngest and oldest candidate

#### ◆ Task 7: Dashboard (Important )

Create a simple Excel dashboard:

- Bar chart → Party vs Total Votes
- Pie chart → Gender Vote Share
- Table → Winners per constituency

#### Tasks (Advanced Level)

- Add a column:  
**Winning Margin = Winner Votes - Runner-up Votes**
- Highlight close contests (< 5000 votes)
- Create slicers in Pivot Tables

#### Real-World Twist

Assume your manager asks:

“Why did XYZ party perform well in urban areas?”

Try to answer using data (even if limited).